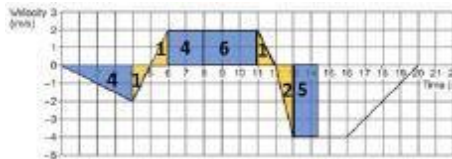


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the steepest gradient. Ignore units. Do not allow -4 .	AO1.1b	B1	4 m s^{-2}
(b)	Shows evidence of determining areas above and below the time axis for values of t between 7 and 9 inclusive Evidence may include values or indication on a diagram	AO3.1b	M1	
	Deduces t_1 value correctly.	AO2.2a	A1	
	Shows evidence of determining areas above and below the time axis for values of t between 13 and 15 inclusive Evidence may include values or indication on a diagram	AO1.1a	M1	
	Deduces t_2 value correctly	AO2.2a	A1	
				Total 5 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds correct acceleration	AO1.1b	B1	0.5 m s^{-2}
(b)	Identifies $5T$ as the distance travelled after the first 15 seconds	AO3.4	B1	Distance at constant speed = $5T$ Distance in first 15 secs =
	Uses the information given to form an equation to find T (award mark for either trapezium expression separate, totalled or	AO3.1b	M1	$\frac{1}{2} \times (3 + 8) \times 10 + \frac{1}{2} \times (8 + 5) \times 5$ $= 55 + 32.5 = 87.5$ $5T + 87.5 = 120$

implied)			So $T = 6.5$
Correctly calculates the distance for the first 15 secs	AO1.1b	A1	
Deduces the values of T from the mathematical models applied	AO2.2a	A1	
			Total 5 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates, with at least one term correct	AO1.1a	M1	$\frac{dv}{dt} = 12t - 36t^2$
	Selects and applies $F = ma$ to 'their' derivative Condone use of 400 for mass	AO1.1a	M1	$F = ma = 0.4 (12t - 36t^2)$
	Obtains correct expression for force FT from 'their' $F = ma$ equation, provided the first M1 has been awarded (may be in factorised form)	AO1.1b	A1F	$= 4.8t - 14.4t^2$
(b)	Integrates v to find r , with at least one term correct	AO3.1b	M1	$r = \int (6t^2 - 12t^3) dt$
	Obtains correct integral (condone absence of c)	AO1.1b	A1	$r = 2t^3 - 3t^4 + c$
	Deduces the value of c using initial conditions FT use of 'their' integral provided M1 awarded	AO2.2a	A1F	When $t = 0$, $r = 0$ so $0 = 2 \times 0^3 - 3 \times 0^4 + c$ so $c = 0$
	Forms and solves an equation for t (condone numerical slip)	AO1.1a	M1	$0 = 2t^3 - 3t^4$ $= t^3(2 - 3t)$ $t = 0$ or $t = \frac{2}{3}$
	Interprets solution, realising that the non-zero time is required (Must include units) FT use of 'their' equation	AO3.2a	A1F	Next at O at $\frac{2}{3}$ seconds

for t provided both M1 marks have been awarded		
Total 8 marks		

Q4.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	80 m s ⁻²
Total 1 mark			

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates two (or four) appropriate distances	AO3.1b	M1	$s_1 = \frac{1}{2} (6 + 10) \times 8 = 64 \text{ m}$
	Obtains correct distances	AO1.1b	A1	$s_2 = \frac{1}{2} \times 10 \times 2 = 10 \text{ m}$
	Obtains correct sum of 'their' distances	AO1.1b	A1F	$s_1 + s_2 = 64 + 10 = 74 \text{ m}$ OR $S_1 = 6 \times 8 = 48 \text{ m}$ $S_2 = \frac{1}{2} \times 4 \times 8 = 16 \text{ m}$ $S_3 = \frac{1}{2} \times 4 \times 2 = 4 \text{ m}$ $S_4 = \frac{1}{2} \times 6 \times 2 = 6 \text{ m}$ $S_1 + S_2 + S_3 + S_4 = 74 \text{ m}$
(b)	Finds difference of 'their' distances from part (a)	AO2.2a	B1F	$64 - 10 = 54 \text{ m}$
(c)	Calculates magnitude of acceleration	AO3.1b	M1	$a_{\max} = \frac{8}{4} = 2$
	Obtains correct resultant force	AO1.1b	A1	$F_{\max} = 800 \times 2 = 1600 \text{ N}$
(d)	Explains that abrupt changes and straight lines in the graph are unlikely in reality	AO3.5b	E1	Change of velocity is unlikely to result in abrupt changes I would expect to see curves on the graph

Total 7 marks

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Integrates both components with at least one correct	AO1.1a	M1	$\mathbf{r} = \int 40e^{-0.2t} dt \mathbf{i} + \int 50(e^{-0.2t} - 1) dt \mathbf{j}$ $= (-200e^{-0.2t} - c) \mathbf{i} + (-250e^{-0.2t} - 50t + d) \mathbf{j}$
	Obtains correct terms. (condone missing constants)	AO1.1b	A1	$t = 0, \mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow c = 200,$ $d = 250$ OR
	Evaluates both constants (or uses definite integration) using 'their' expression for $\mathbf{r}$	AO3.4	M1	$\mathbf{r} = \int_0^t 40e^{-0.2t} dt \mathbf{i} + \int_0^t 50(e^{-0.2t} - 1) dt \mathbf{j}$ $= [-200e^{-0.2t}]_0^t - (-250e^{-0.2t} - 50t) \mathbf{j} \Big _0^t$
	Obtains correct expression	AO1.1b	A1	$\mathbf{r} = 200(1 - e^{-0.2t}) \mathbf{i}$ $+ (250 - 250e^{-0.2t} - 50t) \mathbf{j}$
(b)	Forms equation to find t based on horizontal component	AO3.4	M1	$200(1 - e^{-0.2t}) = 100$ $t = 5 \ln 2 = 3.4657$ $y = 250 - 250e^{-0.2 \times 5 \ln 2} - 50 \times 5 \ln 2$ $= -48.3$
	Obtains correct time	AO1.1b	A1	
	Substitutes 'their' time into vertical component	AO1.1a	M1	
	Obtains correct displacement for 'their' time (to 1 sf, must have metres) FT only if both M1 marks have been awarded	AO3.2a	A1F	The parachutist has a vertical displacement of 50 m below the origin
(c)	Identifies vertical component of the velocity as first step	AO2.4	M1	$\frac{d}{dt} (50(e^{-0.2t} - 1)) = -10e^{-0.2t}$
	Differentiates vertical component of velocity correctly	AO1.1b	A1	As there is no initial vertical component of velocity (and hence no air resistance) the initial acceleration is only due to gravity.
	Considers implication of initial motion and reaches correct conclusion	AO2.2a	R1	Hence g is taken as 10 m s^{-2}
				Total 11 marks

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct horizontal component of the initial velocity	AO1.1b	B1	$2.5U = 40$ $U = 16$
	Forms equation to find vertical component of initial velocity	AO3.3	M1	$-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$
	Obtains correct vertical component of initial velocity	AO1.1b	A1	$V = 8.2625$
	Forms equation for vertical component of velocity at height 3 using 'their' derived values for U and V	AO3.4	M1	$v_y^2 = 8.2625^2 + 2 \times (-9.81) \times 3$
	Obtains correct component of velocity	AO1.1b	A1	$v_y = 3.067...$
	Correct final speed with units, correct for 'their' U and v_y FT applies only if both M1 marks have been awarded	AO3.2a	A1F	$v = \sqrt{16^2 + 3.067^2} = 16.3 \text{ ms}^{-1}$
(b)	States 'their' value of horizontal component of the initial velocity from part (a)	AO3.4	A1F	16 m s^{-1}
(c)	Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning	AO3.5b	E1	It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary... air resistance would slow the ball down, wind might speed the ball up
				Total 8 marks

Q8.

(a) (i) $10^2 = 20^2 + 2 \times a \times 75$

M1: Use of a constant acceleration equation to find a , with $v = 10$ and $u = 20$.

$20^2 = 10^2 + 2 \times a \times 75$ scores M0

A1: Correct equation.

M1A1

$$a = \frac{100 - 400}{150} = -2 \text{ ms}^{-2}$$

A1: Correct acceleration.

A1

For two equation methods award no marks until an equation for a is obtained.

3

(ii) $0 = 20 - 2t$

M1: Using a constant acceleration equation, with $u = 20$ and $v = 0$, to find t using their acceleration from (a) even if positive.

Using $s = 75$ scores M0

M1

$t = 10$ seconds

A1: Correct time from correct working CSO.

A1

2

(iii) $F = 1400 \times 2$

M1: Use of $F = ma$ with $\pm$ their acceleration and mass of 1400.

M1

$= 2800 \text{ N}$

A1F: Correct force. Follow through the magnitude of their acceleration. Answer must be positive. Sign changes do not need to be justified.

A1F

2

(b) $F = 2800 - 200 = 2600 \text{ N}$

B1F: The magnitude of their force minus 200.

Do not award if M1 not awarded in (a)(iii).

Final answer must be positive.

Follow through only if their answer to (a)(iii) is greater than 200.

B1F

1

[8]

Q9.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\mathbf{r}$.

A1: Correct expression for $\mathbf{r}$. Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0 \quad \text{or} \quad t = 30$

M1: Putting their $\mathbf{j}$ component equal to zero to form a quadratic equation.

A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t .

Can be implied by subsequent working in terms of t .

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.

A1: Correct equation.

M1A1

$t = 20$

A1: $t = 20$

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases

If the equation in t in line 2 is not seen:

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and $v = 1.41$

award 4 out of 6

or

then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q10.

(a) $22.4 \sin \theta - 2 \times 9.8 = 0$

M1: Use of $v = u + at$ vertically with

$u = 22.4 \sin \theta$, $v = 0$, $t = 2$ and $a = \pm 9.8$.

A1: Correct equation. (May be in terms of g or contain 9.81..)

M1A1

$$\sin \theta = \frac{19.6}{22.4} = \frac{7}{8} = 0.875$$

A1: Must see either

$$22.4 \sin \theta = 19.6 \text{ or } \frac{19.6}{22.4}$$

AG

A1

Or

$$0 = 22.4 \sin \theta \times 4 - \frac{1}{2} \times 9.8 \times 4^2$$

M1: Use of $s = ut + \frac{1}{2}at^2$ with $u = 22.4 \sin \theta$, $s = 0$, $t = 4$ and $a = \pm 9.8$.

A1: Correct equation.

(M1A1)

$$\sin \theta = \frac{4.9 \times 16}{22.4 \times 4} = 0.875$$

A1: must see $89.6 \sin \theta = 78.4$ or $\frac{78.4}{89.6}$ OE

(A1)

(b) $h_{MAX} = 22.4 \times \sin \theta \times 2 - \frac{1}{2} \times 9.8 \times 2^2$

M1: Using a constant acceleration equation to find height, with $t = 2$, $u = 22.4 \sin \theta$ or 19.6 and $a = \pm 9.8$.

A1: Correct equation.

M1A1

$= 19.6 \text{ m}$

A1: Correct height. AWRT 19.6

Note using $g = 9.81$ gives 19.6, also accept 19.5.

Note: other constant acceleration equations will lead to the same result

A1

Or

$0^2 = (22.4 \times \sin \theta)^2 + 2 \times (-9.8)h_{MAX}$

(M1A1)

$h_{MAX} = 19.6 \text{ m}$

(A1)

(c) $\cos \theta = \frac{\sqrt{15}}{8} = 0.4841$ or $\theta = 61.04^\circ$

B1: Correct value for $\cos \theta$ (accept 0.484) or θ (accept 61.0° or 61° or 1.06 or 1.065 or 1.07 radians). Can be implied.

B1

$AB = 22.4 \times \frac{\sqrt{15}}{8} \times 4 = 43.4 \text{ m}$

M1: Calculation for range with value for $\cos \theta$ and with $t = 4$.

A1F: Correct distance. Follow through incorrect θ . Accept AWRT 43.4 or 43.3 or 43.2.

Do not accept 43.

M1A1F

(d) $22.4 \times (\sin \theta)t - 4.9t^2 = 5$

$\frac{1}{2}$
 M1: Use of $s = ut + \frac{1}{2} at^2$ with correct terms, but not necessarily signs.

M1

$$4.9t^2 - 19.6t + 5 = 0$$

A1: Correct equation.

A1

$$t = 0.274 \text{ or } t = 3.726$$

dM1: Solving their quadratic.

dM1

A1: At least one correct solution. Allow 0.27 or 0.28 and 3.72 or 3.73

A1

$$\text{Time} = 3.726 - 0.274 = 3.45 \text{ seconds}$$

A1: Correct difference. Accept 3.46.

A1

Note: there are other methods which will lead to the correct time:

M1dM1A1 for a constant acceleration equation that gives a time or times from which the final answer can be obtained

A1 Correct time or times

A1 Correct final answer

5

(e) $v_{\min} = 22.4 \times \cos \theta$

M1: Finding horizontal component with candidate's value for $\cos \theta$. Do not award if combined with a non-zero vertical component.

M1

$$= 10.8 \text{ m s}^{-1}$$

A1: Correct speed. Accept 10.9 or 10.85.

A1

Or

$$v_{\min} = \frac{43.4}{4} = 10.9 \text{ to } 3\text{sf}$$

M1: range divided by time of flight

(M1)

A1: Correct speed. Accept 10.9 or 10.85.

(A1)

2

[16]

Q11.

(a) (i) $a = \frac{dv}{dt}$

$= 12t + 8e^{-4t} \text{ m s}^{-2}$

M1 for either term correct

M1A1

2

(ii) When $t = 0.5$, $a = 6 + 8 \times e^{-2}$

m1

$= 7.08 \text{ m s}^{-2}$

Condone 7.07

SC1 for 7.1 with no working

A1

2

(b) Using $F = ma$:

$F = 4 \times 7.08$

$= 28.3 \text{ N}$

Ft from value awarded A1

B1ft

1

(c) $r = \int v dt$

At least two terms correct

M1

$= 2t^3 + \frac{1}{2}e^{-4t} + 8t + c$

Does not need + c

A1

When $t = 0$, $r = 0 \rightarrow c = -\frac{1}{2}$

Does not need $c = -\frac{1}{2}$

m1

$$r = 2t^3 + \frac{1}{2}e^{-4t} + 8t - \frac{1}{2}$$

Need r , s (or words)

A1

4

[9]

Q12.

(a) $s = \frac{1}{2} \times 10 \times 4 + 10 \times 4 + \frac{1}{2} \times (4+7) \times 10 + \frac{1}{2} \times 7 \times 10$

$(= 20 + 40 + 55 + 35)$

M1: Any one term correct.

M1: A second term correct.

A1: Correct expression for total distance.

M1M1A1

$= 150 \text{ m}$

A1: Total distance correct.

A1

OR

$$s = \frac{1}{2} \times (10 + 20) \times 4 + \frac{1}{2} \times (4 + 7) \times 10 + \frac{1}{2} \times 7 \times 10$$

$(= 60 + 55 + 35)$

$= 150 \text{ m}$

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(A1)

OR

$$s = \frac{1}{2} \times 10 \times 4 + 10 \times 4 + 10 \times 4 + \frac{1}{2} \times 10 \times 3 + \frac{1}{2} \times 7 \times 10$$

$(= 20 + 40 + 40 + 15 + 35)$

$= 150 \text{ m}$

(M1M1A1)

(A1)

4

(b) Average Speed = $\frac{150}{40} = 3.75 \text{ m s}^{-1}$

M1: Their total distance divided by 40.

M1

A1F: Correct average speed based on their distance from part (a). Must be correct to three or more significant figures.

A1F

2

(c) $a = \frac{4}{10} = 0.4 \text{ m s}^{-2}$

M1: Any division involving the numbers 10 and 4.

M1

A1: Correct acceleration. CAO

A1

Note on use of constant acceleration equations:
*award M1 for correct equation with correct values
 and A1 for correct final answer.*

2

(d) $F = 200000 \times 0.4 = 80000 \text{ N}$

M1: Multiplication of, 2×10^n , for any integer n , by candidate's acceleration from part (c).

A1F: Correct force based on their answer to part (c) multiplied by 200000.

M1A1F

Note: use of $a = 2.5$ gives 500000 N

Accept 80 kN

2

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[10]

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Q13.

(a) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j})t$

M1: Use of constant acceleration equation to find $\mathbf{v}$ with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$

A1: Correct $\mathbf{v}$.

(Could be done as a column vector.)

M1A1

2

(b) (i) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j}) \times 22.5$

M1: Substitution of 22.5 into their expression for the velocity, even if no marks awarded in part (a).

M1

$$= -5\mathbf{i} + 5\mathbf{j}$$

A1: Correct velocity CAO
(Could be done using column vectors.)

A1

2

(ii) North-west

B1: Correct statement of direction. Accept 315° .
Must follow from correct answer to (b)(i).

B1

1

(c) $(\mathbf{v} =)(4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1: Grouping $\mathbf{i}$ and $\mathbf{j}$ components at some point in the solution.

(Could be done using column vectors.)

Allow $5 = (4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1

$$5^2 = (4 - 0.4t)^2 + (0.5 + 0.2t)^2$$

M1: Seeing both components of their velocity squared and added

A1: Correct equation. (Condone including $\mathbf{i}$ and $\mathbf{j}$.)

For example: $5 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A0

$5^2 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A1

M1A1

$$0.2t^2 - 3t - 8.75 = 0$$

A1: Any correct simplified quadratic equation, with exactly three terms.

A1

$$t^2 - 15t - 43.75 = 0$$

$$t = 17.5 \text{ or } t = -2.5$$

dM1: Solving the quadratic equation.

(Allow one substitution error in correctly quoted formula)

Candidates with an incorrect quadratic equation must show method to get dM1.

dM1

$$t = 17.5$$

A1: Correct positive solution stated.

A1

6

[11]

Q14.

(a) (i) $0.6^2 = 0^2 + 2a \times 0.9$

M1: Correct use of constant acceleration equation with $u = 0$ to find a .

A1: Correct equation.

M1A1

$$a = \frac{0.6^2}{1.8} = 0.2 \text{ ms}^{-2} \quad \mathbf{AG}$$

A1: Correct a but some intermediate working must be seen.

A1

Note that $0^2 = 0.6^2 + 2a \times 0.9$

Scores M0A0A0

Verification methods require a conclusion for full marks to be awarded.

Condone seeing just the second line of working.

3

(ii) $0.9 = \frac{1}{2} (0 + 0.6)t$

M1: Correct use of constant acceleration equation

with $u = 0$ (and $a = 0.2$ if needed) to find t .

M1

$$t = \frac{0.9}{0.3} = 3 \text{ seconds}$$

A1: Correct time.

A1

Note: Do not penalise $0.9 = \frac{1}{2} (0.6 + 0)t$ in the first method.

OR

$$0.6 = 0 + 0.2t$$

(M1)

$$t = \frac{0.6}{0.2} = 3 \text{ seconds}$$

(A1)

Note: $0 = 0.6 + 0.2t$ scores M0A0 in the second method.

OR

$$0.9 = \frac{1}{2} 0.2t^2$$

$$t = 3 \text{ seconds}$$

(M1)

(A1)

2

(b) $T - 800 \times 9.8 = 800 \times 0.2$

M1: Three term equation of motion. Must have these three terms but can have incorrect signs. Must use $a = 0.2$

A1: Correct equation with correct signs.

(Allow 800g)

M1A1

$$T = 7840 + 160 = 8000 \text{ N}$$

A1: Correct tension.

A1

Accept 8008 or 8010 from use of $g = 9.81$.

3

[8]

Q15.

(a) $s = 32 \times 12.5 = 400 \text{ m}$

B1: Correct distance.

B1

1

(b) $1600 = \frac{1}{2} (32 + 18)t$

M1: Seeing 2000 – candidate's answer to part (a) calculated

dM1: Use of constant acceleration equation(s) to find t , with $u = 32$ and $v = 18$

M1dM1

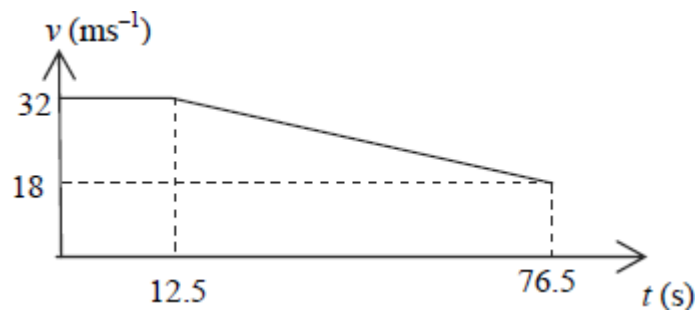
$$t = \frac{1600}{25} = 64 \text{ seconds}$$

A1: Correct time. Accept only 64

A1

3

(c)



B1: Shape of the graph.

B1

B1: Correct velocities (ie 18 and 32) on vertical axis.

B1

B1F: Correct times (ie 12.5 and 76.5) on the horizontal axis.

B1F

(Follow through incorrect answers to part (b)).

Award marks for graph if seen in earlier parts.

3

(d) Average Speed = $\frac{2000}{12.5 + 64} = 26.1 \text{ ms}^{-1}$

M1: Use of 2000 over candidate's total time (not 64 or 12.5).

M1

A1F: Correct speed. AWRT 26.1.

FT candidate's answer to part (b) or (c).

A1F

2

[9]

Q16.

(a) $1000 = V \times 4$

M1: Equation for horizontal motion to find V .

Must not contain g . Could contain $\cos 0^\circ$ or equivalent.

M1

$V = 250 \text{ ms}^{-1}$

A1: Correct V .

A1

2

(b) $(h =) \frac{1}{2} \times 9.8 \times 4^2$

M1: Vertical equation to find height with $u = 0$ and $a = \pm 9.8$.

M1

$= 78.4 \text{ meters to 3sf}$

A1: Correct height. Accept -78.4

A1

2

(c) $(v_y =) 9.8 \times 4 = 39.2 \text{ ms}^{-1}$

M1: Calculation of vertical component of velocity with $u = 0$ and $a = \pm 9.8$.

A1: Correct vertical component.

M1A1

or

$$(v_y =) \sqrt{2 \times 9.8 \times 78.4} = 39.2 \text{ ms}^{-1}$$

$$(v_y =) \sqrt{250^2 + 39.2^2} = 253 \text{ ms}^{-1}$$

dM1: Calculation of speed.

A1: Correct speed.

dM1A1

4

(d) $\tan \alpha = \frac{39.2}{250} \left(\text{or } \tan \alpha = \frac{250}{39.2} \right)$

M1: Using $\tan$ to find angle with opposite and adjacent sides. Can be inverted as shown in brackets.

A1F: Correct trig expression.

M1A1F

$$\alpha = 8.91^\circ$$

A1: Correct angle.

A1

OR

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$$\sin \alpha = \frac{39.2}{253} \left(\text{or } \sin \alpha = \frac{250}{253} \right)$$

M1: Using $\sin$ to find angle with hypotenuse and one other side. Can be changed as shown in brackets.

A1F: Correct trig expression.

(M1A1F)

$$\alpha = 8.91^\circ$$

A1: Correct angle.

(A1)

OR

$$\cos \alpha = \frac{250}{253(.055)} \left(\text{or } \cos \alpha = \frac{39.2}{253} \right)$$

M1: Using cos to find angle with hypotenuse and one other side. Can be changed as shown in brackets.

A1F: Correct trig expression.

(M1A1F)

$$\alpha = 8.91^\circ$$

A1: Correct angle. Accept 8.83° from this method.

(A1)

Note: Accept 8.98° from 253.1

Accept negative angles

Note: FT value of V from (a) and speed from (c) if needed. Do not FT 39.2 from (c) in place of 253.

Note: Accept energy methods if used correctly in part (c).

3

[11]

Q17.

(a) $\mathbf{v} = (0.5\mathbf{i} + 0.375\mathbf{j}) \times 20 (=10\mathbf{i} + 7.5\mathbf{j})$

M1: Calculating velocity with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$ and $t = 20$.

A1: Correct expression for velocity.

M1A1

$$v = \sqrt{10^2 + 7.5^2} = 12.5 \text{ ms}^{-1}$$

dM1: Calculating speed.

A1: Correct speed.

dM1A1

4

(b) $\tan \theta = \frac{0.5}{0.375} \text{ or } \frac{10}{7.5} \left(\text{or } \tan \theta = \frac{0.375}{0.5} \text{ or } \frac{7.5}{10} \right)$

M1: Using trig to find angle. Can be inverted as shown in brackets.

A1F: Correct trig expression with any correct equivalent fraction.

M1A1F

$$\theta = 053^\circ$$

A1: Correct angle to the nearest degree. Accept 53°.

A1

OR

$$\cos \theta = \frac{7.5}{12.5} \text{ or } \frac{0.375}{0.625} \left(\text{or } \cos \theta = \frac{10}{12.5} \right)$$

(M1A1F)

$$\theta = 053^\circ$$

Note: For 37° award M1A0A0
But for $90 - 37 = 53^\circ$ award M1A1A1.
For 127° , award M1A1A0

(A1)

OR

$$\sin \theta = \frac{10}{12.5} \text{ or } \frac{0.5}{0.625} \left(\text{or } \sin \theta = \frac{7.5}{12.5} \right)$$

Note: 53.1° as final answer scores M1A1A0

(M1A1F)

$$\theta = 053^\circ$$

Condone finding angle from acceleration or position vector.

(A1)

3

(c) $(\mathbf{r} =) \frac{1}{2} (0.5\mathbf{i} + 0.375\mathbf{j}) t^2 (= 0.25 t^2 \mathbf{i} + 0.1875 t^2 \mathbf{j})$

M1: Finding an expression for position vector in terms of t .

A1: Correct position vector.

M1A1

$$500^2 = (0.25 t^2)^2 + (0.1875 t^2)^2$$

dM1: Using distance to form an equation for t .

A1: Correct equation.

dM1A1

$$t = 4 \sqrt{\frac{500^2}{0.25^2 + 0.1875^2}} = 40 \text{ seconds}$$

A1: Correct time.

A1

OR

$$a = 0.625$$

M1: Finding magnitude of acceleration.

A1: Correct acceleration

(M1A1)

$$500 = \frac{1}{2} 0.625 t^2$$

dM1: Using distance to form an equation for t .

A1: Correct equation.

(dM1A1)

$$t = 40$$

A1: Correct time.

(A1)

OR

$$400 = \frac{1}{2} 0.5t^2 \text{ or } 300 = \frac{1}{2} 0.375t^2$$

M1: Working with one component.

A1: Correct distance (300 or 400)

A1: Correct equation.

(M1A1)

(A1)

$$t^2 = 1600$$

dM1: Solving for t .

(dM1)

$$t = 40$$

A1: Correct t .

Note: $500 \div 12.5 = 40$ is not acceptable and scores 0

(A1)

5

[12]

Q18.

EXAM PAPERS PRACTICE

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(a) $\mathbf{a} = \frac{dv}{dt}$

$$\mathbf{a} = -8e^{-2t} \mathbf{i} + (6 - 6t)\mathbf{j}$$

M1: Differentiating with either of the two components correct. Do not need to see i or j .

A1: Correct i component.

A1: Correct j component.

M1

A1

A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$
 $\mathbf{F} = 5 \times \{-8e^{-2t} \mathbf{i} + (6 - 6t)\mathbf{j}\}$

$$= -40e^{-2t} \mathbf{i} + (30 - 30t)\mathbf{j}$$

M1: Multiplying their acceleration by 5, even if not a vector.

A1: Correct expression.

M1
A1

2

(ii) Magnitude of $\mathbf{F}$ is

$$\{(-40)^2 + (30)^2\}^{\frac{1}{2}}$$

M1: Finding magnitude from two non-zero terms. Must add terms and square root. Condone $\{(40)^2 + (30)^2\}^{\frac{1}{2}}$

M1

$$= 50$$

A1: Correct answer only.

In this part, condone lack of negative signs in expression for force in (b) (i).

A1

2

(c) When $\mathbf{F}$ acts due west, j component is zero

$$30 - 30t = 0$$

M1: Putting j component equal to zero.

M1

$$t = 1$$

A1: Correct time.

A1

2

(d) $\mathbf{r} = -2e^{-2t} \mathbf{i} + (3t^2 - t^3) \mathbf{j} + \mathbf{c}$

M1: Integration with either of the two components correct. Do not need to see $\mathbf{i}$ or $\mathbf{j}$.

A1: Correct $\mathbf{i}$ component.

A1: Correct $\mathbf{j}$ component.

Condone lack of $+\mathbf{c}$

M1

A1

A1

$$\text{When } t = 0, \mathbf{r} = 6\mathbf{i} + 5\mathbf{j} \therefore \mathbf{c} = 8\mathbf{i} + 5\mathbf{j}$$

dM1: Finding $\mathbf{c}$ using $6\mathbf{i} + 5\mathbf{j}$ and $e^0 = 1$.

dM1

$$\therefore \mathbf{r} = (8 - 2e^{-2t}) \mathbf{i} + (5 + \frac{1}{4}t^4 - t^3) \mathbf{j}$$

A1: Correct position vector.

A1

5

[14]

Q19.

(a) $0 = 4.9 - 9.8t$

M1: Use of constant acceleration equation to find t with $v = 0$.

A1: Correct equation

M1A1

$$t = \frac{4.9}{9.8} = 0.5 \text{ s}$$

A1: Correct time

A1

3

(b) $0^2 = 4.9^2 - 2 \times 9.8s$

M1: Use of constant acceleration equation to find s with $v = 0$.

M1

$$s = \frac{4.9^2}{19.6} = 1.225 \text{ m}$$

A1: Correct height.

A1

$$H_{\max} = 1.225 + 1 = 2.23 \text{ m}$$

A1: Correct total height.

A1

3

(c) 1 second

B1: Correct time.

B1

1

(d) 4.9 m s^{-1}

B1: Correct speed.

B1

1

[8]

Q20.

(a) $\mathbf{v} = 4.5\mathbf{i} + 2\mathbf{j}$

B1: Correct velocity

B1
1

(b) $\mathbf{v} = 4.5\mathbf{j}$

B1: Correct velocity

B1
1

(c) Due North

B1: Correct direction

B1
1

(d) $4.5\mathbf{j} = 4.5\mathbf{i} + 2\mathbf{j} + 5\mathbf{a}$

M1: Use of a constant acceleration equation to find a

A1: Correct equation

M1A1

$$\mathbf{a} = \frac{-4.5\mathbf{i} + 2.5\mathbf{j}}{5} = -0.9\mathbf{i} + 0.5\mathbf{j}$$

A1: Correct acceleration

A1
3

(e) $\mathbf{F} = 250(-0.9\mathbf{i} + 0.5\mathbf{j}) = -225\mathbf{i} + 125\mathbf{j}$

M1: Using Newton's second law.

M1

$$F = \sqrt{225^2 + 125^2} = 257 \text{ N}$$

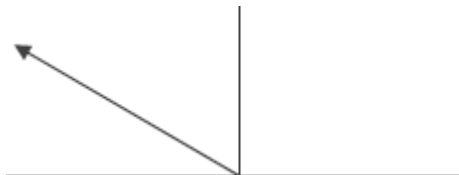
M1: Finding magnitude.

A1: Correct magnitude of the force

M1A1

3

(f)



B1: Diagram with arrow in the correct quadrant.

B1

$$\tan^{-1}\left(\frac{125}{225}\right) = 29.1^\circ$$

M1: Use of trig to find an angle.

M1

A1: Correct angle from working.

A1

OR

$$\tan^{-1}\left(\frac{0.5}{0.9}\right) = 29.1^\circ$$

$$180 - 29.1 = 151^\circ$$

A1: Finding required angle correctly.

A1

4

[13]

Q21.

(a) $5 = \frac{1}{2} \times 9.8t^2$

M1: Equation based on vertical motion with no velocity component, with ± 5 and ± 9.8

M1

A1: Correct equation

A1

$$t = \sqrt{\frac{5}{4.9}} = 1.01 \text{ s}$$

AG

A1: Correct time from correct working.

Must see square root or $t^2 = 1.02$ OE

Note: If $g = 9.81$ is used for the first time deduct one mark. Should still get 1.01 seconds.

A1

3

(b) $15 = V \times \sqrt{\frac{5}{4.9}}$

M1: Using distance = speed $\times$ time OE

M1

$$V = 15 \sqrt{\frac{4.9}{5}} = 14.8$$

*A1: Correct speed.
 Accept AWRT 14.8 or 14.9.
 Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 14.9 m s^{-1} from $g = 9.81$.*

A1

2

(c) $v_v = \pm 9.8 \times \sqrt{\frac{5}{4.9}} (= \pm 9.899)$

*M1: Calculating vertical component of velocity.
 A1: Correct value. Accept 9.9 or similar*

M1A1

or

$$v_v = \sqrt{2 \times 9.8 \times 5} = 9.899$$

$$v = \sqrt{9.899^2 + 14.8^2} = 17.8 \text{ m s}^{-1}$$

dM1: Finding magnitude (with addition not subtraction of squares inside the square root).

dM1

*A1: Correct speed.
 Accept AWRT 17.8 or AWRT 17.9.
 Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 17.9 m s^{-1} from $g = 9.81$*

A1F

4

(d) $\tan \alpha = \frac{9.899}{14.8} \text{ or } \frac{14.8}{9.899}$

$\alpha = 34^\circ$

M1: Use of one of trig equations shown.

M1

A1F: Anything which rounds to 34° or 56°

A1F

A1F: 34° CAO (33° scores M1A1A0)

A1F

3

$$\sin \alpha = \frac{9.899}{17.8} \text{ or } \frac{14.8}{17.8}$$

$\alpha = 34^\circ$

Only follow through if all method marks in (b) and (c) have been awarded (except the dM if tan used).

(M1)(A1F)(A1F)

$$\cos \alpha = \frac{14.8}{17.8} \text{ or } \frac{9.899}{17.8}$$

$$\alpha = 34^\circ$$

(M1)(A1F)(A1F)

[12]

Q22.

(a) $10 = 0 \times 2.5 + \frac{1}{2} a \times 2.5^2$

M1: Use of constant acceleration equation to find a with u = 0.

M1

*A1: Correct equation.
NOTE: If v is found first, do not award any marks for part (a) until an equation to find a is produced. This could be from graphical method or from the use of*

$$s = \frac{1}{2} (u + v)t.$$

A1

$$a = \frac{20}{2.5^2} = 3.2 \text{ m s}^{-2}$$

A1: Correct acceleration

A1

3

(b) $10 = \frac{1}{2} (0 + v) \times 2.5$

M1: Use of constant acceleration equation to find v with u = 0.

M1

$$v = 8 \text{ m s}^{-1}$$

A1: Correct speed from correct working.

A1

OR

$$10 = \frac{1}{2} v \times 2.5$$

$$v = 8 \text{ m s}^{-1}$$

NOTE: If v is found in part (a), with correct working award full marks.

OR

$$v^2 = (0^2 +) 2 \times 3.2 \times 10$$

NOTE: Accept $3.2 \times 2.5 = 8$

$$v = 8 \text{ m s}^{-1}$$

OR

$$v = (0 +) 3.2 \times 2.5 = 8 \text{ m s}^{-1}$$

AG

2

(c) $t = \frac{90}{8} = 11.25 \text{ s}$

B1: Calculation of correct additional time. Could be implied by later working.

B1

$$\begin{aligned} \text{Total Time} &= 2.5 + 11.25 \\ &= 13.75 \\ &= 13.8 \text{ s} \end{aligned}$$

M1: Addition of their time for the 90 metres and the 2.5 seconds.

M1

A1: Correct total time. Accept 13.75.

A1

NOTE: $22.5 + 2.5 = 25$ scores B0M1A0

3

(d) $\frac{100}{13.75} = 7.27 \text{ m s}^{-1}$

M1: Finding average speed. Must see 100 and their answer from part (c).

M1

A1F: Follow through candidate's time from part (c), regardless of working in part (c). Allow 7.25 m s^{-1} from 13.8 seconds.

A1F

2

Q23.

(a) $v^2 = 0^2 + 2 \times 9.8 \times 15$

M1: Use of constant acceleration equation to find v with $u = 0$ and $a = \pm 9.8$.

M1

$v^2 = 294$

A1: Correct equation

A1

$v = 17.1 \text{ m s}^{-1}$

A1: Correct speed from correct working. Accept AWRT 17.1. Accept 17.15.

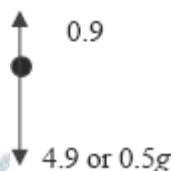
Accept $7\sqrt{6}$

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 17.2 m s^{-1} from $g = 9.81$.

A1

3

(b) (i)



B1: Correct diagram, with arrows and labels. Must see 0.9 and 4.9 or 0.5g (or 4.905 if working with $g = 9.81$).

B1

1

(ii) $4.9 - 0.9 = 0.5a$

M1: Uses $0.5a$.

B1: Explicit statement of " $4.9 - 0.9$ " or " $mg - 0.9$ " or " $0.5g - 0.9$ ".

M1B1

$(a =) \frac{4}{0.5} = 8 \text{ ms}^{-2}$ **AG**

A1: Correct acceleration from correct working. Can be awarded without the B1 mark.

Must see $\frac{4.9(\text{or } 0.5g) - 0.9}{0.5}$ or $\frac{4}{0.5}$ or $4 = 0.5a$

A1

Note: If $g = 9.81$ is used candidates will get 8.01 m s^{-2} . Deduct 1 mark if 8.01 is seen.

Examples:

$4.9 = 0.5a + 0.9$ M1 B0 A0

$a = 8$

$4 = 0.5a$ M1 B0 A1

$a = 8$

If candidates only write

$a = \frac{4}{0.5} = 8$ award M0 B0 A0.

3

(iii) $v^2 = 0^2 + 2 \times 8 \times 15$

M1: Use of constant acceleration equation to find v with $u = 0$ and $a = \pm 8$.

M1

$v = 15.5 \text{ m s}^{-1}$

A1: Correct speed from correct working.

Accept AWRT 15.5 or truncated to 15.4.

Accept $4\sqrt{15}$.

A1

2

- (iv) The air resistance force will not be constant, but changes **as the speed of the ball changes** (or **changes as the ball accelerates**).

B1: Correct explanation, key words in bold.

Do not award mark for statements that imply that the acceleration causes the air resistance to change.

B1

1

[10]

Q24.

(a) $(8\mathbf{i} + 12\mathbf{j}) + (4\mathbf{i} - 4\mathbf{j}) = 12\mathbf{i} + 8\mathbf{j}$

M1: Adding forces to find resultant.

A1: Correct resultant force.

M1A1

(b) $4\mathbf{a} = 12\mathbf{i} + 8\mathbf{j}$ or $(\mathbf{a} =) \frac{12\mathbf{i} + 8\mathbf{j}}{4}$

M1: Use of Newton's second law with $4\mathbf{a}$ and their answer to part (a).

M1

$(\mathbf{a} =) 3\mathbf{i} + 2\mathbf{j}$ **AG**

A1: Correct acceleration from correct equation.

A1

(c) (i) $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + (3\mathbf{i} + 2\mathbf{j}) \times 20$
 $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + 60\mathbf{i} + 40\mathbf{j}$

B1: Seeing $60\mathbf{i} + 40\mathbf{j}$ or $(3\mathbf{i} + 2\mathbf{j}) \times 20$

B1

M1: Use of constant acceleration equation with $t = 20$ and $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j}$.

M1

$\mathbf{v} = (40\mathbf{i} + 32\mathbf{j}) - (60\mathbf{i} + 40\mathbf{j})$ **AG**
 $= -20\mathbf{i} - 8\mathbf{j}$

*A1: Correct velocity from correct working, with one of the intermediate lines of working (or equivalent) shown.
 Note: Candidates may use $\mathbf{u}$ instead of $\mathbf{v}$ in their working.*

Example:

Starting with

$\mathbf{v} = 40\mathbf{i} + 32\mathbf{j} + (3\mathbf{i} + 2\mathbf{j}) \times 20$

Scores B1M1A0.

A1

Note on Verification Method:

$\mathbf{v} = (-20\mathbf{i} - 8\mathbf{j}) + (3\mathbf{i} + 2\mathbf{j}) \times 20$ **B1M1**

$= (-20 + 60)\mathbf{i} + (-8 + 40)\mathbf{j}$

$= 40\mathbf{i} + 32\mathbf{j}$ **A1**

Similarly, verification to confirm acceleration from the two velocities is acceptable.

(ii) $(\mathbf{v} =) (-20\mathbf{i} - 8\mathbf{j}) + (3\mathbf{i} + 2\mathbf{j})$

B1: Correct velocity vector.

Note " $\mathbf{v} =$ " does not need to be seen.

B1

(iii) $(\mathbf{v} =) (3t - 20)\mathbf{i} + (2t - 8)\mathbf{j}$

M1: Velocity vector seen split into components. Condone omission of $\mathbf{i}$ and $\mathbf{j}$
Note: This can be implied by later working, such as the second line of this solution.

M1

$$(3t - 20)^2 + (2t - 8)^2 = 8^2$$

dM1: Equation based on speed of 8.

dM1

A1: Correct unsimplified equation.

A1

$$13t^2 - 152t + 400 = 0$$

A1: Simplified quadratic equation

A1

$$t = \frac{152 \pm \sqrt{152^2 - 4 \times 13 \times 400}}{2 \times 13}$$

dM1: Solving quadratic equation, to obtain two solutions.

dM1

$$t = 4 \text{ or } t = 7.69$$

A1: Both correct solutions. Accept

$$\text{AWRT } 7.7 \text{ or } 7.6 \text{ or } \frac{100}{13}$$

Note: Using calculator to solve quadratic is acceptable.

A1

6

[14]

Q25.

(a) $5 = \frac{1}{2} \times 9.8t^2$

M1: Equation based on vertical motion with no velocity component, with ± 5 and ± 9.8

M1

A1: Correct equation

A1

$$t = \sqrt{\frac{5}{4.9}} = 1.01 \text{ s} \quad \mathbf{AG}$$

*A1: Correct time from correct working.
 Must see square root or $t^2 = 1.02$ OE
 Note: If $g = 9.81$ is used for the first time deduct one mark. Should still get 1.01 seconds.*

A1

3

(b) $15 = v \times \sqrt{\frac{5}{4.9}}$

M1: Using distance = speed $\times$ time OE

M1

$$v = 15 \sqrt{\frac{4.9}{5}} = 14.8$$

*A1: Correct speed.
 Accept AWRT 14.8 or 14.9.
 Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 14.9 m s^{-1} from $g = 9.81$.*

A1

2

(c) $v_y = \pm 9.8 \times \sqrt{\frac{5}{4.9}} (= \pm 9.899)$

*M1: Calculating vertical component of velocity.
 A1: Correct value. Accept 9.9 or similar*

M1A1

or

$$v_y = \sqrt{2 \times 9.8 \times 5} = 9.899$$

$$v = \sqrt{9.899^2 + 14.8^2} = 17.8 \text{ m s}^{-1}$$

dM1: Finding magnitude (with addition not subtraction of squares inside the square root).

dM1

*A1: Correct speed.
 Accept AWRT 17.8 or AWRT 17.9.*

A1F

Note: If $g = 9.81$ is used for the first time deduct one mark. Should get 17.9 m s^{-1}

from $g = 9.81$

4

(d) $\tan \alpha = \frac{9.899}{14.8} \text{ or } \frac{14.8}{9.899}$
 $\alpha = 34^\circ$

M1: Use of one of trig equations shown.

M1

A1F: Anything which rounds to 34° or 56°

A1F

A1F: 34° CAO (33° scores M1A1A0)

A1F

Only follow through if all method marks in (b) and (c) have been awarded (except the dM if tan used).

3

$\sin \alpha = \frac{9.899}{17.8} \text{ or } \frac{14.8}{17.8}$
 $\alpha = 34^\circ$

(M1)(A1F)(A1F)

$\cos \alpha = \frac{14.8}{17.8} \text{ or } \frac{9.899}{17.8}$
 $\alpha = 34^\circ$

(M1)(A1F)(A1F)

(e) Particle

B1: Particle assumption

B1

Experiences no air resistance **or** no wind **or** only gravity
or no other forces acting **or** no spin.

*B1: Other assumption.
 Ignore any other assumptions.*

B1

2

[14]

Q26.

(a) $12g - T = 12a$

M1: Three term equation of motion, with

$12g$ (or 117.6), $12a$ (not $12ga$) and T .
 A1: Correct equation

M1A1

$$T - 8g = 8a$$

M1: Three term equation of motion, with
 $8g$ (or 78.4), $8a$ (not $8ga$) and T .
 A1: Correct equation

M1A1

$$4g = 20a$$

$$a \left(= \frac{4g}{20} \right) = 1.96 \text{ m s}^{-2} \quad \text{AG}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who
 consistently use signs in the opposite
 direction throughout, provided they
 give their final answer as 1.96. If final
 answer is -1.96 don't award final A1 mark.

Special Case:

Whole String Method $4g = 20a$ = and

$$a = \frac{4g}{20} = 1.96$$

OE M1A1A1

5

(b) $T = 8g + 8 \times 1.96 = 94.1 \text{ N}$

M1: Use of three term equation of motion
 to find T , with $a = 1.96$.

A1: Correct tension.

Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation
 to find v , with $a = 1.96$ and $u = 0$.

A1: Correct v .

Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation
 to find v , with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

$$v = 9.68 \text{ m s}^{-1}$$

A1F: Correct v . FT initial velocity from (c)(i).
For example 11.8 from 7.84.

A1F

3

Use of $g = 9.81$

(a) 1.962

M1A1M1A1A0

(b) 94.2

M1A1

(c) (ii) 9.69

M1A1A1

[12]

Q27.

(a) $\mathbf{v} = 9\mathbf{i} + 7\mathbf{j}$

B1: Correct initial velocity.

B1

1

(b) $\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$

M1: Writing $\mathbf{v}$ in the form $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$$= (9\mathbf{i} + 7\mathbf{j}) + (-0.01\mathbf{i} - 0.03\mathbf{j})t$$

$$\mathbf{a} = -0.01\mathbf{i} - 0.03\mathbf{j}$$

A1: Correct acceleration.

A1

2

(c) $9 - 0.01t = -(7 - 0.03t)$

M1: Equation involving both $\mathbf{i}$ and $\mathbf{j}$ components of their velocity.

A1: Correct equation, from their acceleration.

M1A1

$$t = \frac{16}{0.04} = 400$$

A1: Correct time, from their acceleration.

A1

3

[6]

Q28.

(a) $14.7 \sin \alpha - 9.8t (= 0)$

M1: Equation for vertical velocity being zero at highest point. Must have $\sin \alpha$ with ± 9.8 .

A1: Correct equation.

M1A1

$$t = \frac{14.7 \sin \alpha}{9.8} = \frac{3 \sin \alpha}{2} \quad \mathbf{AG}$$

A1: Correct result from correct working.

A1

OR

$$14.7 \sin \alpha T - 4.9T^2 (= 0)$$

$$T = \frac{14.7 \sin \alpha}{4.9} = 3 \sin \alpha$$

$$t = \frac{3 \sin \alpha}{2}$$

All marks awarded for last line, from correct working.

(M1)(A1)(A1)

3

(b) (i) $7 = 14.7 \sin \alpha \left(\frac{3 \sin \alpha}{2} \right) - 4.9 \left(\frac{3 \sin \alpha}{2} \right)^2$

M1: Expression including vertical displacement at height 7, using expression from part (a) and with $\pm g$ or equivalent.

M1

A1: Correct expression.

A1

$$7 = 11.025 \sin^2 \alpha$$

dM1: Simplified expression with $\sin^2 \alpha$.

dM1

dM1: Finding an angle. Must have previous dM1 mark.

dM1

$$\alpha = \sin^{-1} \left(\sqrt{\frac{7}{11.025}} \right) = 52.8^\circ$$

*A1: Correct angle.
Accept 52.7°, 52.9°.*

A1

OR

$$0^2 = (14.7 \sin \alpha)^2 + 2 \times (-9.8) \times 7$$

(M1)(A1)

$$\sin^2 \alpha = \frac{2 \times 9.8 \times 7}{14.7^2}$$

(dM1)

$$\alpha = 52.8^\circ$$

(dM1)(A1)

5

(ii) $OA = 14.7 \cos 52.8^\circ \times 3 \sin 52.8^\circ$

B1: Use of $3 \sin \alpha$ with their α .

M1: Finding horizontal displacement.

including $14.7 \cos \alpha$ with $3 \sin \alpha$ or $\frac{3 \sin \alpha}{2}$.

B1M1

$$OA = 21.2 \text{ m}$$

A1: Correct distance. Accept 21.3 m.

A1

3

Use of $g = 9.81$:

(a) M1A1A0

(b) (i) 52.8° or 52.9° M1A1dM1dM1A1

(b) (ii) 21.2 B1M1A1

[11]

Q29.

(a) 30 seconds

B1: Correct statement of time.

B1

1

(b) $s_1 = \frac{1}{2} \times 40 \times 20 = 400 \text{ m}$

M1: A method for calculating the first

distance. Must see 40 and $\frac{1}{2}$.

M1

A1: Correct distance.

A1

OR

$$s_1 = \frac{1}{2} \times (20 + 0) \times 40 = 400 \text{ m}$$

(M1)(A1)

OR

$$a = -\frac{20}{40} = -\frac{1}{2}$$

$$0^2 = 20^2 + 2\left(-\frac{1}{2}\right)s$$

$$s = 20^2 = 400 \text{ m}$$

Note on third method: Must see $-\frac{1}{2}$ or $-\frac{20}{40}$ plus attempt to find distance for M1.

(M1)(A1)

2

(c) $s_2 = \frac{1}{2} \times 50 \times 20 = 500 \text{ m}$

M1: Method for finding the second distance and calculating the total distance.

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OR

$$s_2 = \frac{1}{2} \times (0 + 20) \times 50 = 500 \text{ m}$$

(M1)

OR

$$a = \frac{20}{50} = \frac{2}{5}$$

$$20^2 = 0^2 + 2\left(\frac{2}{5}\right)s$$

$$s = 20^2 \times \frac{5}{4} = 500 \text{ m}$$

Note on third method: Must see $\frac{2}{5}$ or $\frac{20}{50}$ plus attempt to find distance.

(M1)

Total = $400 + 500 = 900 \text{ m}$

A1F: Correct total distance. Award the follow through mark for correct addition of 500 and their answer to (b).

A1F

2

(d) $v_{\text{AVERAGE}} = \frac{900}{120} = 7.5 \text{ ms}^{-1}$

M1: Their total distance divided by 120

M1

A1F: Correct average speed based on their answer to (c).

A1F

2

(e) $120 \times 20 - 900 = 1500 \text{ m}$

M1: Multiplication of 20 and 120 to find distance.

Note: Award M1 if 2400 seen in this part.

A1F: Correct difference based on their answer to (c) provided final answer is positive.

M1A1F

2

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[9]

Q30.

(a) $12g - T = 12a$

M1: Three term equation of motion, with $12g$ (or 117.6), $12a$ (not $12ga$) and T .

A1: Correct equation

M1A1

$T - 8g = 8a$

M1: Three term equation of motion, with $8g$ (or 78.4), $8a$ (not $8ga$) and T .

A1: Correct equation

M1A1

$$4g = 20a$$

$$a \left(= \frac{4g}{20} \right) = 1.96 \text{ m s}^{-2} \quad \text{AG}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 1.96. If final answer is – 1.96 don't award final A1 mark.

Special Case:

Whole String Method $4g = 20a$ and

$$a = \frac{4g}{20} = 1.96 \quad \text{OE M1A1A1}$$

5

(b) $T = 8g + 8 \times 1.96 = 94.1 \text{ N}$

M1: Use of three term equation of motion to find T , with $a = 1.96$.

A1: Correct tension.

Accept 94.08.

M1A1

2

(c) (i) $v = 0 + 1.96 \times 2 = 3.92 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v , with $a = 1.96$ and $u = 0$.

A1: Correct v .

Using $s = 4$ scores M0.

M1A1

2

(ii) $v^2 = 3.92^2 + 2 \times 9.8 \times 4$

M1: Use of constant acceleration equation to find v , with $a = \pm 9.8$ and $u \neq 0$.

M1

A1F: Correct equation. FT initial velocity from (c)(i).

A1F

$$v = 9.68 \text{ m s}^{-1}$$

A1F: Correct v . FT initial velocity from (c)(i). For example 11.8 from 7.84.

A1F

3

(iii) $4 = \frac{1}{2}(-3.92 + 9.68)t$

M1: Use of $s = \frac{1}{2}(u + v)t$

M1

A1: Correct values.

A1

A1: Correct signs.

A1

$t = 1.39$

dM1: Solving for t.

dM1

A1: Correct t.

A1

OR

$-4 = 3.92t - 4.9t^2$

$4.9t^2 - 3.92t - 4 = 0$

$t = \frac{3.92 \pm \sqrt{3.92^2 - 4 \times 4.9 \times (-4)}}{2 \times 4.9}$

M1: Forming a quadratic with candidates u from (c)(i) or v from (c)(ii) with 4.9 or 9.8.

(M1)

A1: Correct terms in quadratic.

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(A1)

A1: Correct signs in quadratic.

(A1)

$t = 1.39$ or $t = -0.588$

dM1: Solving quadratic (do not penalise for negative discriminant).

(dM1)

$t = 1.39$

A1: Correct root seen (other root does not need to be seen).

(A1)

OR

$$t_{\text{up}} + t_{\text{down}} = 0.4 + 0.4 + 0.588$$

M1: Finding total time from two or three times.

(M1)

A1: 0.4 or 0.8 seen.

(A1)

dM1: Finding second or third time for downward motion.

(dM1)

A1: Obtaining 0.588 or 0.988.

(A1)

$$= 1.39 \text{ (to 3SF)}$$

A1: 1.39. Accept 1.38.

(A1)

OR

$$9.68 = -3.92 + 9.8t$$

M1: Use of $v = u + at$

(M1)

A1: Correct values.

(A1)

A1: Correct signs.

(A1)

$$t = \frac{13.6}{9.8} = 1.39$$

dM1: Solving for t

(dM1)

A1: Correct t

(A1)

5

Use of $g = 9.81$

- | | |
|----------------|-------------|
| (a) 1.962 | M1A1M1A1A0 |
| (b) 94.2 | M1A1 |
| (c) (ii) 9.69 | M1A1A1 |
| (c) (iii) 1.39 | M1A1A1dM1A1 |

[17]

Q31.

(a) $14.7 \sin \alpha - 9.8t (= 0)$

*M1: Equation for vertical velocity being zero at highest point. Must have $\sin \alpha$ with ± 9.8 .
A1: Correct equation.*

M1A1

$$t = \frac{14.7 \sin \alpha}{9.8} = \frac{3 \sin \alpha}{2} \quad \mathbf{AG}$$

A1: Correct result from correct working.

A1

OR

$$14.7 \sin \alpha T - 4.9 T^2 (= 0)$$

$$T = \frac{14.7 \sin \alpha}{4.9} = 3 \sin \alpha$$

$$t = \frac{3 \sin \alpha}{2}$$

All marks awarded for last line, from correct working.

(M1)(A1)(A1)

3

(b) (i) $7 = 14.7 \sin \alpha \left(\frac{3 \sin \alpha}{2} \right) - 4.9 \left(\frac{3 \sin \alpha}{2} \right)^2$

M1: Expression including vertical displacement at height 7, using expression from part (a) and with $\pm g$ or equivalent.

M1

A1: Correct expression.

A1

$$7 = 11.025 \sin^2 \alpha$$

dM1: Simplified expression with $\sin^2 \alpha$.

dM1

dM1: Finding an angle. Must have previous dM1 mark.

dM1

$$\alpha = \sin^{-1} \left(\sqrt{\frac{7}{11.025}} \right) = 52.8^\circ$$

*A1: Correct angle.
Accept 52.7° , 52.9° .*

A1

OR

$$0^2 = (14.7 \sin \alpha)^2 + 2 \times (-9.8) \times 7$$

(M1)(A1)

$$\sin^2 \alpha = \frac{2 \times 9.8 \times 7}{14.7^2}$$

(dM1)

$$\alpha = 52.8^\circ$$

(dM1)(A1)

5

(ii) $OA = 14.7 \cos 52.8^\circ \times 3 \sin 52.8^\circ$

B1: Use of $3 \sin \alpha$ with their α .

M1: Finding horizontal displacement.

including $14.7 \cos \alpha$ with $3 \sin \alpha$ or $\frac{3 \sin \alpha}{2}$.

B1M1

$$OA = 21.2 \text{ m}$$

A1: Correct distance. Accept 21.3 m.

A1

3

(c) Ball is a particle/No spin.

B1: Particle assumption.

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B1

No air resistance/No wind/Constant acceleration of 9.8/Only force is weight.

B1: Air resistance assumption.

B1

2

[13]

Use of $g = 9.81$:

(a)

M1A1A0

(b) (i) 52.8° or 52.9°

M1A1dM1dM1A1

(b) (ii) 21.2

B1M1A1

Q32.

(a) Using $\mathbf{F} = m\mathbf{a}$,

$$400 \cos \frac{\pi}{2} t \mathbf{i} + 600t^2 \mathbf{j} = 200 \mathbf{a}$$

M1

$$\mathbf{a} = 2 \cos \frac{\pi}{2} t \mathbf{i} + 3t^2 \mathbf{j}$$

A1

2

(b) $\mathbf{v} = \int \mathbf{a} \, dt$

M1 for either $\int \mathbf{a} \, dt$ or 1 of 2 terms correct

M1

$$= \frac{4}{\pi} \sin \frac{\pi}{2} t \mathbf{i} + t^3 \mathbf{j} + \mathbf{c}$$

m1 for + c

A1m1

When $t = 4$, $\mathbf{r} = -3\mathbf{i} + 56\mathbf{j}$,
 $64\mathbf{j} + \mathbf{c} = -3\mathbf{i} + 56\mathbf{j}$

m1

$$\therefore \mathbf{c} = -3\mathbf{i} - 8\mathbf{j}$$

$$\therefore \mathbf{v} = \left(\frac{4}{\pi} \sin \frac{\pi}{2} t - 3 \right) \mathbf{i} + (t^3 - 8) \mathbf{j}$$

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Do not accept $\frac{2}{\pi}$ Accept 1.27 for $\frac{4}{\pi}$

A1

5

(c) When particle is moving due west, northerly component is zero

M1

$$\therefore t^3 - 8 = 0$$

A1ft

$$t = 2$$

A1

3

- (d) When $t = 2$, $\mathbf{v} = -3\mathbf{i} + 0\mathbf{j}$

B1ft

Speed of particle is 3 m s^{-1}

B1 for change -3 to $+3$

B1

2

[12]

Q33.

- (a) $11g - T = 11a$

M1: Equation of motion for A, containing T, 11g or 107.8 and 11a.

M1

A1: Correct equation

A1

$$T - 9g = 9a$$

M1: Equation of motion for B containing T, 9g or 88.2 and 9a.

M1

A1: Correct equation

A1

$$2g = 20a$$

AG

$$a = 0.98 \text{ m s}^{-2}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is -0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and

$$a = 2g / 20 = 0.98 \text{ OE M1A1A1}$$

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

- (b) (i) $v = 0 + 0.98 \times 0.5 = 0.49 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225$ m

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$$0.49^2 = 0^2 + 2 \times 0.98s$$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and their v .

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$$d = 2 \times 0.1225$$

M1: Doubling distance or use of $d/2$ in their original equation.

M1

$$= 0.245 \text{ m}$$

*A1: Correct final distance. Allow 0.244 or 0.246.
(Use of $0.5 \times 0.49 = 0.245$ scores zero unless justified)*

A1

4

[11]

Q34.

(a) $0^2 = (30 \sin 35^\circ)^2 + 2 \times (-9.8)s$

M1: Equation to find the max height, with $v = 0$

M1

A1: Correct equation

A1

M1: Solving for the height

M1

$$s = \frac{(30 \sin 35^\circ)^2}{2 \times 9.8} = 15.1 \text{ m}$$

A1: Correct height

A1

4

(b) $28^2 = (30 \cos 35^\circ)^2 + v_y^2$

M1: Equation to find vertical component

M1

A1: Correct equation

A1

$$v_y = \sqrt{28^2 - (30 \cos 35^\circ)^2}$$

M1: Solving equation

M1

$$= 13.4198 = 13.4 \text{ m s}^{-1} \quad \text{AG}$$

A1: Correct speed from correct working.

A1

4

(c) $\tan \theta = \frac{13.4}{30 \cos 35^\circ}$

M1: Expression to find the angle.

M1

$$\theta = 28.6^\circ$$

A1: Correct angle.

A1

2

[10]

Q35.

- (a) Peg is smooth

B1: Correct assumption

B1

1

- (b) String is light

B1: First correct assumption

B1

String is inextensible or inelastic

B1: Second correct assumption

B1

Tension is the same throughout the string

Note: Ignore any additional assumptions.

2

- (c) $11g - T = 11a$

M1: Equation of motion for A, containing, 11g or 107.8 and 11a.

M1

A1: Correct equation

A1

$$T - 9g = 9a$$

M1: Equation of motion for B containing, 9g or 88.2 and 9a.

M1

A1: Correct equation

A1

$$2g = 20a$$

AG

$$a = 0.98 \text{ m s}^{-2}$$

A1: Correct acceleration from correct working.

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they give their final answer as 0.98. If final answer is - 0.98 don't award final A1 mark.

Special Case:

Whole String Method $2g = 20a$ and

$$a = 2g/20 = 0.98 \text{ OE M1A1A1}$$

Use of $g = 9.81$ gives 0.981. If this is the first time award M1A1M1A1A0, but don't penalise again on the same script.

5

(d) (i) $v = 0 + 0.98 \times 0.5 = 0.49 \text{ m s}^{-1}$

M1: Use of constant acceleration equation to find v with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct v

A1

2

(ii) $s = 0 + \frac{1}{2} \times 0.98 \times 0.5^2 = 0.1225 \text{ m}$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and $t = 0.5$.

M1

A1: Correct distance. Accept 0.122 or 0.123

A1

OR

$$0.49^2 = 0^2 + 2 \times 0.98s$$

M1: Finding distance travelled by each particle with $u = 0$, $a = 0.98$ and their v

(M1)

$$s = \frac{0.49^2}{2 \times 0.98} = 0.1225$$

A1: Correct distance. Accept 0.122 or 0.123

(A1)

$$d = 2 \times 0.1225$$

M1: Doubling distance or use of $d/2$ in their original equation.

M1

$$= 0.245 \text{ m}$$

A1: Correct final distance. Allow 0.244 or 0.246. (Use of $0.5 \times 0.49 = 0.245$ scores zero unless justified)

A1

***If candidates calculate the distance first
award marks as above (see (d) (i)) or:***

***M1: Use of constant acceleration equation
to find v with $u = 0$, $a = 0.98$ and $s = 0.1225$.***

A1: Correct v

***Note: If parts (i) and (ii) are not
separated or clearly labelled still award
marks for both parts if justified.***

[14]



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